Fertile ^{232}Th can be transformed into fissile ^{233}U in any thermal reactor. The reactions in ^{232}Th irradiated by neutrons are given in Fig. 20.3. Of the thermally fissile atoms ^{233}U has the highest $\sigma_f/\sigma_{n,\gamma}$ ratio, i.e. highest fission efficiency. The η-value is high enough to permit breeding in the thermal region. Capture of neutrons in ^{233}U is not a serious drawback as a second capture (in ^{234}U) yields fissile ^{235}U, but reduces the breeding gain because two neutrons are consumed without a net increase in fissile material. Since σ_{tot} increases with neutron energy (from ~7 b at 0.025 eV to ~26 b for reactor conditions; cf. also Fig. 19.3) the slightly harder neutron spectrum in PHWR and GCR make such reactors the prime candidates for a thorium-uranium fuel cycle together with the molten salt reactor (§20.3). The fuel may be arranged in a core (^{233}U) and blanket (^{232}Th) fashion, or mixed fissile and fertile material as, for example, in the HTGR prototype (High Temperature Gas-cooled Reactor) graphite matrix fuels, or as a metal fluoride melt. The initial ^{233}U must be produced from thorium in reactors fuelled with ^{235}U (or ^{239}Pu) or in special accelerator driven devices. After sufficient amounts of ^{233}U have been produced, the Th-U fuel cycle may become self sustaining, i.e. thermal breeding is established.

The advantage of the Th—U fuel cycle is that it increases nuclear energy resources considerably because thorium is about three times more abundant on earth than uranium and almost as widely distributed. In combination with the uranium fuel cycle it could more than double the lifetime of the uranium resources by running the reactors at a high conversion rate (~1.0) and recycling the fuel. Very rich thorium minerals are more common than rich uranium minerals. The presence of extensive thorium ores has motivated some countries (e.g. India) to develop the Th—U fuel cycle.

No full-scale Th-U fuel cycle has yet been demonstrated and reprocessing has only been demonstrated on an experimental scale. The fuel cycle has to overcome the high activity problems due to the presence of ^{228}Th formed in the thorium fraction and ^{232}U formed in the uranium fraction (Fig. 20.3). The Th—U fuel cycle has a rather specific advantage over the U—Pu cycle in that its high active waste from reprocessing contains a much smaller amount of long-lived heavy actinides, although it probably contain some ^{231}Pa. With regard to nuclear weapons proliferation ^{233}U is almost as good a weapons material as ^{239}Pu and easier to produce as a single isotope by continuous withdrawal of protactinium, since it is the decay product of ^{233}Pa ($t_{½}$ 27 d), see also §21.7.

21.6. REPROCESSING OF URANIUM AND MIXED OXIDE FUELS

The main purpose of commercial reprocessing is:

(1) to increase the available energy from fissile and fertile atoms;

(2) to reduce hazards and costs for handling the high level wastes.

Two other reasons are sometimes mentioned:

(3) to reduce the cost of the thermal reactor fuel cycle;

(4) to extract valuable byproducts from the high active waste.

The 30% savings in natural uranium for LWR and similar reactors and the hundredfold energy resource expansion for FBRs when reprocessing spent fuel, has already been discussed in the previous section. The economic advantage of reprocessing depends on the cost and availability of natural ("yellow cake") uranium, on enrichment and other front end activities, and on the prevailing energy price (mainly based on fossil fuels). At present, cheap uranium is abundant.

However, as mentioned before, older aluminium canned fuel needs either to be treated in order to cover it by a thick and resistant oxide film, or to be reprocessed, in order to avoid severe corrosion of the can and release of fuel constituents into the storage facility.

The reprocessing plants at La Hague, France, have a total capacity of ~1600 tons IHM/y, the Magnox reprocessing plant and the THORP (THermal Oxide Reprocessing Plant) at Sellafield, UK, have capacities of ~1500 and ~850 tons IHM/y, respectively. A commercial reprocessing plant under construction in Japan is designed to have a capacity of 800 tons IHM/y. In Siberia, a large russian reprocessing plant is under construction. A number of smaller, older plants are also in operation in several countries.

Figure 21.12 is a schematic representation of reprocessing of spent LWR fuel. The main steps are: (i) the *head end* section, in which the fuel is prepared for chemical separation; (ii) the main fractionation (partitioning) of U, Pu, and FP; (iii) purification of uranium; (iv) purification of plutonium; (v) waste treatment; and (vi) recovery of chemicals. These steps are described in the following sections.

21.6.1. Head end plant

Figure 21.13 is a simplified drawing of one of the French oxide fuel head end plants at La Hague. The flasks with the used fuel assemblies are lifted by a crane into water-filled pits, where the flasks are unloaded and decontaminated. The assemblies are stored for a desired time and then transferred to a shielded dismantling and chopping section. Some BWR fuels have end parts, which can be mechanically dismantled; this is not the case for PWR fuels, in which the end parts have to be cut off. The fuel pins are cut into pieces 3–5 cm long, either under water or in air. At THORP, UK, and in the newest part of the La Hague plant, dry charging and chopping is used.

The chopping is usually achieved with a shearing knife (cutter), but other techniques for removing or opening up the zircaloy (or stainless steel) cans have been tried. Previously, chemical decanning was used at some plants, e.g. Hanford and Eurochemic, but such techniques increase the amounts of active waste considerably.

The chopped pieces are transferred to the dissolver unit, where the oxide fuel is leached by boiling in 6–11 M HNO_3 (the cladding hulls do not dissolve) in thick stainless steel vessels provided with recirculation tubes and condenser. The hulls are measured for residual uranium or plutonium, and, if sufficiently clean of fissile material,

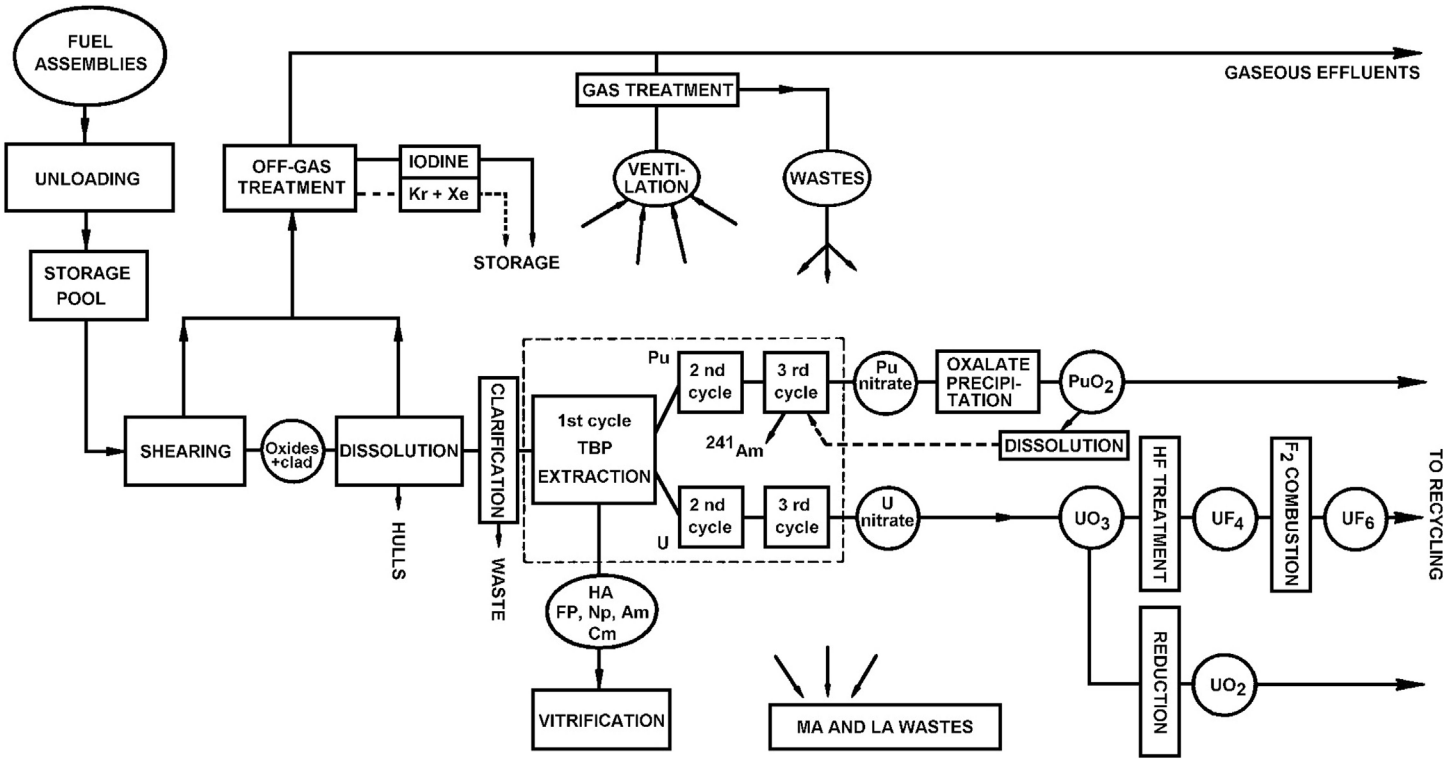


Figure 21.12 Schematic representation of the reprocessing of spent LWR fuel. *(From Musikas and Schulz.)*

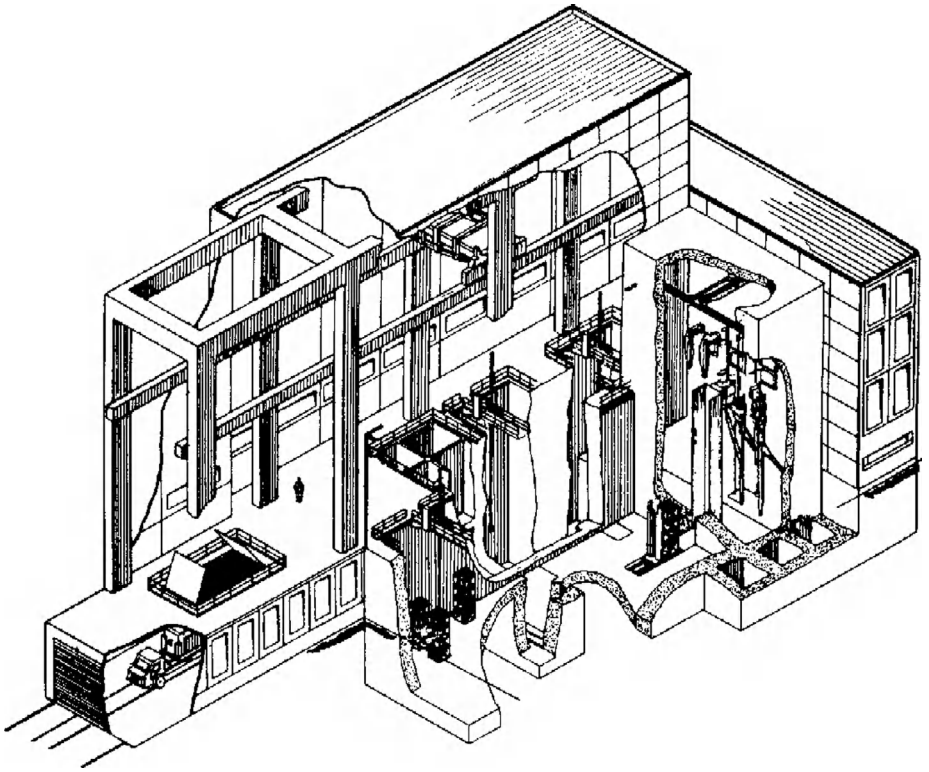


Figure 21.13 Head end oxide fuel building at La Hague, France.

are discharged to the waste treatment section of the plant. To improve the dissolution, some fluoride ($\leq$0.05 M AlF_3) may be added to the HNO_3. The F^- forms strong complexes with some metal ions such as zirconium, while its corrosion of the stainless steel equipment has been found to be negligible. Soluble poisons, such as cadmium nitrate or gadolinium nitrate, are often added to the nitric acid to assure the criticality control of the dissolution operation.

High burnup of fissile material leads to a high fission product content in the fuel elements resulting in the formation of seminoble metal fission product alloys, which are insoluble in boiling nitric acid. The insoluble material consists of mm sized metal particles of Ru, Rh, Tc, Mo, and Pd, the larger ones often called "ingots". These metal particles usually contain negligible amounts of uranium and plutonium and can be filtered as high level solid waste, HLSW.

When the fuel pins are cut, the volatile fission products contained in the gas space between the fuel oxide pellets and the canning is released (mainly ^{85}Kr, ^{3}H, ^{129}I, and ^{131}I). These gases are ducted to the dissolver off-gas treatment system. Gases released during dissolution are Kr, Xe, I_2, T_2, THO, RuO_4, CO_2, minor amounts of fission

product aerosols, and large amounts of H_2O, HNO_3, and nitrogen oxides. Oxygen or air is fed into the off-gas stream to allow recovery of part of the nitrogen oxides. The overall dissolution stoichiometry is

$$UO_2 + 2HNO_3 + \tfrac{1}{2}O_2 \rightarrow UO_2(NO_3)_2 + H_2O \qquad (21.10)$$

The gas streams pass to a condenser which reclaims and returns some nitric acid to the dissolver. The noncondensibles are discharged to the off-gas treatment system.

When the dissolution is completed, the product solution is cooled and transferred to the input measurement–clarification (filter and/or centrifuge) feed adjustment unit. At this point the uranium is in the hexavalent state, and plutonium in the tetravalent.

21.6.2. Separation methods

The specifications for purified uranium and plutonium to be recycled are summarized in Table 21.7. Comparing these data with those presented before shows that at $t_{cool} \sim 1$ y the fission product activity must be reduced by a factor of $\sim 10^7$ and the uranium content in plutonium by a factor of $\sim 2 \times 10^4$. The large number of chemical elements involved (FPs, actinides *and* corrosion products) make the separation a difficult task. Additional complications arise from radiation decomposition and criticality risks and from the necessity to conduct all processes remotely in heavily shielded enclosures under extensive

Table 21.7 Specifications for reprocessed uranium and plutonium

Uranyl nitrate:
- Uranium concentration 1–2 M
- Free HNO_3 ≥1 M
- Impurities:
  - Fe, Cr, Ni ≤500 ppm
  - Boron equivalents ≤8 ppm[†]
- Fission products ≤19 MBq/kg U
- α-activity (excluding uranium) ≤250 kBq/kg U

Plutonium nitrate:
- Plutonium concentration ~1 M
- Free HNO_3 2–10 M
- Impurities:
  - Metallic ≤5000 ppm
  - Uranium ≤5000 ppm
  - Boron equivalents ≤10 ppm
  - Sulfate ≤1000 ppm
- Fission products ($t_{½}$ > 30 d) ≤1.5 GBq/kg Pu[‡]
- ^{241}Am content (9 months after delivery to MOX-plant) ≤5000 ppm

[†]The equivalent values are B 1.0, Cd 0.4, Gd 4.4, Fe 0.0007, etc. The amount measured for each of these elements multiplied by the factor indicated must not be more than 8 ppm.
[‡]^{95}Zr − Nb ≤ 185 MBq/kg Pu.
(From IAEA 1977)

health protection measures. As a result reprocessing is one of the most complicated chemical processes ever operated on an industrial scale.

The problem encountered by the chemists of the Manhattan Project in the 1940's was the selection of satisfactory separation techniques. Advantage was taken of the relative stability of the oxidation state of uranium (+6) and most fission products, and the redox lability of plutonium (+3, +4, and +6). In the earliest process, only plutonium was isolated by precipitating plutonium in the reduced state as PuF_3 or PuF_4 together with all insoluble FP fluorides. This was followed by a second stage dissolving the precipitate, oxidizing plutonium to the +6 state, and a new fluoride precipitation, leaving relatively pure plutonium in the supernatant. In a final step plutonium was again reduced and precipitated as fluoride. This principle was used for the first isolation of hundreds of kilograms of plutonium at the Hanford Engineering Works, USA, with phosphate precipitation of Pu(+3) and Pu(+4), but not of Pu(+6) which does not form an insoluble phosphate. Since Bi^{3+} was used as a carrier for the precipitate, it is referred to as the *bismuth phosphate process*.

This principle of oxidizing and reducing plutonium at various stages of the purification scheme has been retained in all subsequent processes. No other element has the same set of redox *and* chemical properties as plutonium, though some elements behave as Pu^{3+} (e.g. the lanthanides), some like Pu^{4+} (e.g. zirconium) and some like PuO^{+} (e.g. uranium). Numerous redox agents have been used, e.g. $K_2Cr_2O_7$ (to PuO^{+}), $NaNO_2$ (to Pu^{4+}), hydrazine, ferrous sulfamate, and U^{4+} (to Pu^{3+}), cf. Table 14.4

The precipitation technique is not suitable for large-scale, continuous remote operations in which both uranium and plutonium have to be isolated in a very pure state from the fission products. It was therefore replaced in the late 1940's by *solvent extraction* in which the fuels were dissolved in nitric acid and contacted with an organic solvent which selectively extracted the desired elements. The technique has been mentioned in §§14.2.4.3, 17.7.4, and 18.4.3 but is described in more detail in Appendix A.

The first solvent to be adopted at Hanford was methyl isobutyl ketone ("MIBK" or "Hexone"). This solvent forms *adduct compounds* with coordinatively unsaturated compounds like the actinide nitrates, e.g. $Pu(NO_3)_4S_2$, where S represents the adduct molecule

$$Pu^{4+} + 4NO_3^- + 2S(org) \rightarrow Pu(NO_3)_4S_2(org) \qquad (21.11)$$

The corresponding adduct compounds for 3- and 6-valent actinides are $An(NO_3)_3S_3$ and $AnO_2(NO_3)_2S_2$. These chemically saturated neutral compounds are soluble to different extent in organic solvents like kerosene, and — in the case of hexone — by hexone itself. The process using hexone is referred to as the *Redox process*.

In the United Kingdom, β,β'-dibutoxydiethylether ("dibutyl carbitol" or "Butex") was selected as organic solvent; it forms the same kind of adduct compounds as hexone. Though more expensive, it was more stable, less flammable and gave better separations.

Many other similar solvent systems, as well as *chelating agents*, have been tested. Thenoyltrifluoroacetone (HTTA) was found to form strong complexes with the actinides (e.g. $Pu(TTA)_4$, $UO_2(TTA)_2$), which show very high distribution ratios in favour of organic solvents. Though useful in the laboratory they were not found suitable for large scale commercial nuclear fuel reprocessing. One of the most useful recent extraction agents for actinide separation is di-2-ethylhexylphosphoric acid (HDEHP) which has found several industrial uses outside the nuclear energy industry, e.g. separation of rare earth elements.

A drawback of hexone and butex is the need to use salting-out agents (salts like $Al(NO_3)_3$ added to the aqueous phase) in order to obtain sufficiently high extraction factors. Such salt additions increase the liquid waste volumes. Further, hexone was unstable at high nitric acid concentrations. All this led to the search for a better extractant.

Presently, tributyl phosphate (TBP) is the extractant used in all reprocessing plants. It acts as an adduct former and is normally used as a 30% solution in kerosene. It forms the basis for the *PUREX process* (Plutonium Uranium Redox EXtraction). TBP is cheaper than Butex, more stable, less flammable, and gives better separations. The limitation to 30% in TBP is needed in order to prevent such an increase in density of the organic phase, due to extracted uranium and plutonium, that it becomes equally dense as the aqueous phase or even denser[2].

Other extractants, especially tertiary amines, have been tested for some steps in reprocessing. The amines form organic soluble complexes with negatively charged metal complexes (used in uranium purification, §14.2.4.2). The use in the basic PUREX cycle of a secondary extractive reagent can improve the decontamination factor[3].

As alternatives to the aqueous separation processes, "dry" techniques have also been studied, but none has been used on an industrial scale. Examples are the following:

(a) *Halide volatility.* Many FP and the high valency actinides have appreciable vapor pressures; this is particularly true for the fluorides. In *fluoride volatilization* the fuel elements are dissolved in a molten fluoride salt eutectic ($NaF + LiF + ZrF_4$, 450°C) in the presence of HF. The salt melt is heated in F_2, leading to the formation of UF_6, which is distilled; it may be possible also to distill PuF_6, though it is much less stable. The process has encountered several technical difficulties.

(b) *Molten salt extraction.* The fuel is dissolved as above or in another salt melt. With a heat resistant solvent of low volatility (e.g. 100% TBP), actinides and FP distribute themselves between the two phases analogous to solvent extraction. This technique is of interest for continuous reprocessing of molten salt reactor fuel or partitioning in an accelerator driven transmuter (§20.3.1). An advantage is the higher radiation resistance of the nonaqueous system.

[2] In one process developed for the steel industry, 75% TBP was used.

[3] The decontamination factor is defined as (concentration before separation)/(concentration after separation).

(c) *Molten salt transport*. The fuel is dissolved in a metallic melt, e.g. a molten Cu-Mg alloy, which is in contact through a stirred molten chloride salt at about 800°C with another metallic melt containing a reductant, e.g. a molten Zn-Mg alloy. Noble metal FPs are retained in the Cu-containing melt, whereas U and/or Pu is collected in the Zn-containing melt. The less noble FPs concentrate in the molten salt.

(d) *Molten salt electrorefining*. The spent fuel acts anode in a molten salt also containg a pure metal cathode. By applying an electric field between the anode and cathode, material is dissolved at the anode and deposited at the cathode. Careful control of the applied potential makes it possible to obtain an extremely pure cathode product. The process was initially developed for purification of plutonium metal alloys proposed as fuel for the Los Alamos Molten Plutonium Reactor Experiment (LAMPRE).

(e) *Molten metal purification*. Metallic fuel elements can be molten and/or dissolved in molten metals (e.g. a zinc alloy). In the presence of (deficient amounts of) oxygen, strongly electropositive fission elements form oxides, which float to the surface of the melt as slag and can thus be removed, while volatile FPs distill. The residual melt would mainly contain U, Pu, Zr, Nb, Mo, and Ru ("fissium alloy") and can be reused in new metallic fuel elements. This melt refining technique has been tested on metallic breeder reactor fuel elements (in EBR II). Molten chlorides have also been used to remove americium from molten plutonium metal scrap.

21.6.3. PUREX separation scheme

The distribution of uranium, plutonium, and some FPs between 30% TBP (in kerosene) and aqueous solutions of varying HNO_3 concentration is shown in Figure 14.9. D_{Sr} and D_{Cs} are $\ll 0.01$ for all HNO_3 concentrations. The distribution of the trivalent lanthanides and actinides fall within the Eu-Am area. Many fission products (most I, II, III, V, and VII-valent species) are not extracted, i.e. $D \leq 0.01$; see Appendix A. Thus at high HNO_3 concentration Pu(IV), Pu(VI), and U(VI) are extracted but very little of the FPs. At low HNO_3 concentration the D-value for actinides of all valency states is $\ll 1$, and consequently the tetra- and hexavalent actinides are stripped from the organic phase by dilute HNO_3. This chemistry is the basis for the PUREX process.

The PUREX process is presented schematically in Figure 21.12, where the solvent extraction steps are within the dotted frame. Three purification cycles for both uranium and plutonium are shown. High levels of beta and gamma activity is present only in the first cycle, in which >99% of the fission products are separated. The principle of the first cycle is shown in Fig. 21.14. The two other cycles are based upon the same chemical reactions as in the first cycle; the purpose is to obtain additional decontamination and

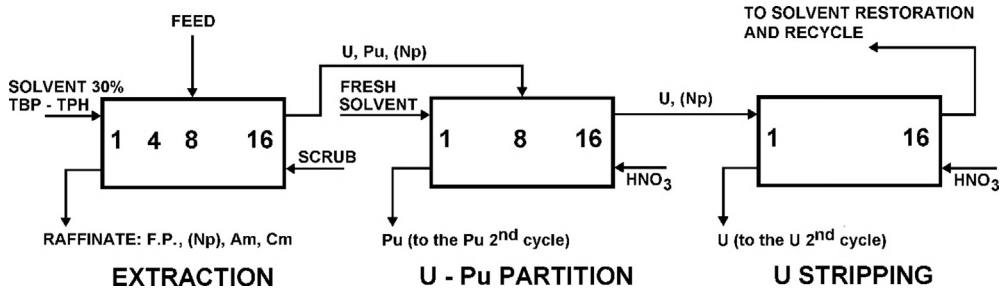


Figure 21.14 Flowsheet of the first purification cycle in the PUREX process; TPH, tetrapropylene, is a commercial dodecane. *(From Musikas and Schulz.)*

overall purity of the uranium and plutonium products. Each square in Figure 21.14 indicates a number of solvent extraction stages of the particular equipment used: pulsed columns, mixer-settlers, etc. (see Appendix A).

In the first cycle > 99.8% U and Pu (in VI and IV state, respectively) are co-extracted from 3–4 M HNO_3 into the kerosene-TBP phase, leaving >99% of the FP in the aqueous raffinate. In the *partitioning* stage, plutonium is reduced to the III state by a solution containing a suitable reductant, e.g. U(IV) nitrate or Fe(II) sulfamate; the plutonium is stripped to a new aqueous phase and transferred to the plutonium purification section. Uranium, which as U(VI) (and U(IV)) stays in the organic phase, is stripped by dilute HNO_3 in a third stage. After concentration by evaporation it is sent to the uranium purification section.

The uranium purification contains two extraction-stripping stages. Plutonium tracers are removed by a reducing agent, e.g. U(IV):

$$U^{4+} + 2Pu^{4+} + 2H_2O \rightarrow UO^{+} + 2Pu^{3+} + 4H^{+} \qquad (21.12)$$

U(IV) is preferred over Fe(II)-sulfamate in some plants as it avoids the introduction of foreign substances. The final, concentrated uranium solution may be percolated through a column filled with silica gel, which removes residual FP, particularly Zr–Nb.

The plutonium purification may be achieved by additional TBP extraction cycles. U(IV) cannot be used as reductant in this part of the process. The final uranium and plutonium products are nitrate solutions whose conversion to oxides, fluorides, etc., have partly been described earlier (§21.1).

The chemical problems encountered in the solvent extraction are:

(i) The choice of diluent: A mixture of aliphatic hydrocarbons, and occasionally pure dodecane, are most common; improper choice of diluent may lead to formation of a third liquid phase, slow extraction kinetics, difficult phase disengagement (i.e. separation of the organic and aqueous phases in the extraction equipment), etc.

(ii) The choice of reductant for the reaction Pu(IV) → Pu(III): The use of U(IV) as reductant for Pu(IV) introduces new uranium in the streams; Fe(II) sulfamate ($Fe(SO_3NH_2)_2$) adds objectionable inorganic salts to the aqueous high level waste; hydroxylamine (NH_2OH) is a kinetically slow-reducing agent; electrolytic reduction of Pu(IV) to Pu(III) is lacking experience.

(iii) Solvent degradation: Radiation decomposes TBP into lower phosphates and butyl alcohol. The main products are dibutyl phosphate (DBP, $(BuO)_2POOH$) and monobutyl phosphate (MBP, $BuOPO(OH)_2$) which form strong complexes with many of the fission products as well as plutonium. As these radiolysis products are formed the decontamination efficiency decreases and losses of fissile material to the aqueous waste streams increase. The solvent is treated to remove the degradation products prior to recycle in the process, e.g. by washing the TBP solution successively with Na_2CO_3, NaOH, and dilute acid solutions.

To reduce some of these difficulties and to generally improve the efficiency of the U and Pu extraction (leaving a waste almost free of these elements) as well as the decontamination factors, other extractants than TBP have been suggested, for example dialkylamides, R-CON-R'$_2$ where R may be C_3H_7 and R' is $CH_2CHC_2H_5C_4H_9$, "DOBA". The advantages of this type and other proposed alternatives to TBP are that they are completely incinerable (while TBP leaves a phosphate waste), the radiolytic products are not deleterious for the process performance, and that no reducing agent is necessary to partition uranium and plutonium. Finally, the ash from combustion could be leached and its content of actinides recovered and returned to the process — further reducing the content of these elements in intermediate level waste.

During the decontamination steps, acid streams containing small amounts of actinides and fission products are produced. These streams are evaporated to concentrate the metal ions and recycle them. Nitric acid is recovered from the condensates and recycled. Excess HNO_3 may be destroyed by formaldehyde. Fission product concentrates are routed to the aqueous raffinate from the first extractor of the partitioning cycle which contains >99% of the FP. This constitutes the *high level liquid waste* (HLLW, or alternatively called HAW, *high active waste*). All other liquid wastes can be subdivided into *intermediate level waste.*

(ILW or MLW) or *low level waste* (LLW). It is an important goal in all reprocessing operations to reduce the amount of intermediate and low level waste streams as far as possible and to route most of what remains to the HLLW stream. In order to reduce volume the HLLW is often concentrated further by evaporation. The concentrate is usually designed as HLC. Waste treatment is discussed in §21.8 .

As we have seen above the completely dominating recycling of nuclear fuel today is based on the recovery of uranium and plutonium through the PUREX process. However, in a more sustainable future nuclear system recycling of essentially all the actinides will be desirable. Due to their abundance in the used nuclear fuel the main

focus has been on recovering neptunium, americium and curium from the streams in, or out from, the PUREX process. Neptunium can be recovered by adjustment of the redox properties of the initial extraction stages. Then it will follow uranium to the uranium purification cycle, where it can be separated from U and recovered separately. The main problem encountered, however, is that both the heavy actinides as well as the lanthanides are trivalent and of similar size making an Am-Cm separation from the PUREX waste stream difficult.

Two main strategies exist to perform this separation of the trivalent actinides from the lanthanides. Both techniques start by extracting both the actinides and the lanthanides from the PUREX raffinate but then the routes differ. In the technique utilised by e.g. the reversed TALSPEAK (Trivalent Actinide − Lanthanide Separation by Phosphorous reagent Extraction from Aqueous Komplexes) process in the US the actinides are selectively stripped from the mixture extracted by HDEHP in while in the European techniques exemplified by the SANEX (Selective Actinide Extraction) processes the actinides are selectively extracted from the mixture. Both these techniques have been tested successfully using genuine spent nuclear fuel.

In recent times, however, questions have been raised about proliferation issues using the PUREX process considering the fact that one of the results is a pure plutonium stream. This makes the safe guard issues in relation to PUREX based processes complicated. Instead it has been suggested to extract all the actinides together from the dissolved used nuclear fuel. Such a process is called a GANEX (Group Actinide Extraction) process. This extraction is performed in two steps. First the bulk uranium is extracted and in the next step all the other actinides (and remaining uranium) is extracted. Such a process opens for the possibility of a homogeneous recycling where essentially the new nuclear[1] fuel is produced directly from the waste of the used one utilising a similar composition where as much uranium as needed can be added. Naturally the use of this fuel types will require the use of fast neutron spectrum reactors.

21.6.4. Engineering aspects and operation safety

All operations in a reprocessing plant have to cope with the necessity of preventing nuclear criticality and of protecting operations personnel and the environment from exposure to or contamination by radioactivity. Thus all equipment has dimensions which are safe against criticality, as, for example, annular or pipe shaped tanks for liquid storage instead of conventional tanks, or are provided with neutron absorbers. All equipment is made of stainless steel and is installed in concrete cells with wall thicknesses up to 2 m at the head end, partitioning, and waste treatment sections. All operations are carried out in airtight enclosures at reduced pressure relative to working areas.

In the event of failure of equipment within a radioactive area, three courses of action may be taken: (i) switching to duplicate equipment, (ii) replacing or repairing equipment

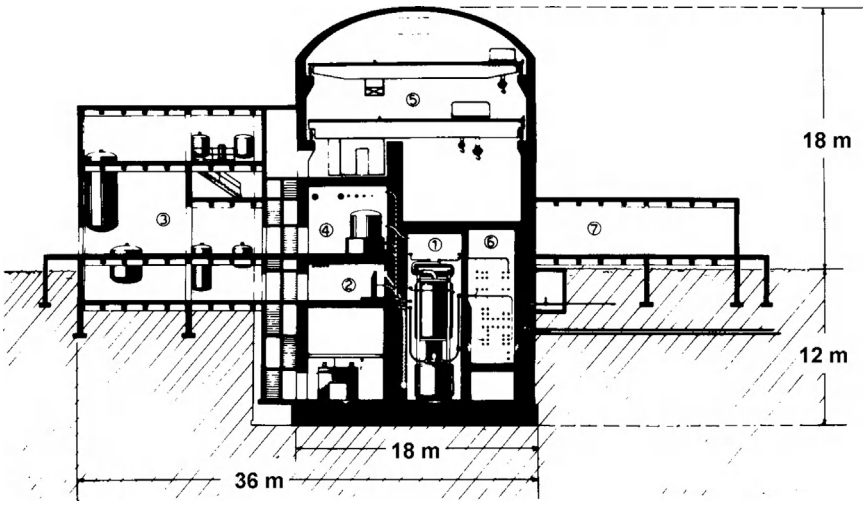


Figure 21.15 Principle of a reprocessing plant with remote maintenance. 1, canyon with process equipment; 2, control room; 3, feed preparations; 4, feed input; 5, cranes; 6, piping; 7, analytical section.

by remote methods; *remote maintenance*, or (iii) repairing by *direct maintenance* after decontamination.

Plants were originally constructed for either completely remote maintenance or completely direct maintenance. Figure 21.16 illustrates the principle of a plant for remote maintenance. All equipment is installed in a large canyon with piping in a parallel corridor. The equipment can be replaced by an overhead crane operated from a shielded room; malfunctioning equipment could be transferred to a decontamination and repair unit or to an equipment "grave". With this philosophy conventional type chemical plant equipment can be used, though redesigned for the remote replacement. Of particular importance in such plants are good joints for piping, electricity, etc., for remote connection.

The British Nuclear Fuels Plc. (BNFL) Magnox and THORP plants at Sellafield, UK, are designed for no maintenance, which in practice might mean either direct maintenance or remote maintenance; Fig. 21.16. All kinds of common reasons for failure in chemical plants have been eliminated or minimized by using welded joints and no moving parts. Thus there are no leaking fittings, frozen valves, or stuck pumps. All welding is carefully controlled by ultrasound or radiography. Active liquids are transported by gas (usually steam) jets or lifts, or by vacuum. Liquid levels and volumes are measured by differential pressure gages and by weighing. Samples for analysis are remotely withdrawn and analyzed in shielded boxes or glove boxes depending on their activity outside the enclosure.

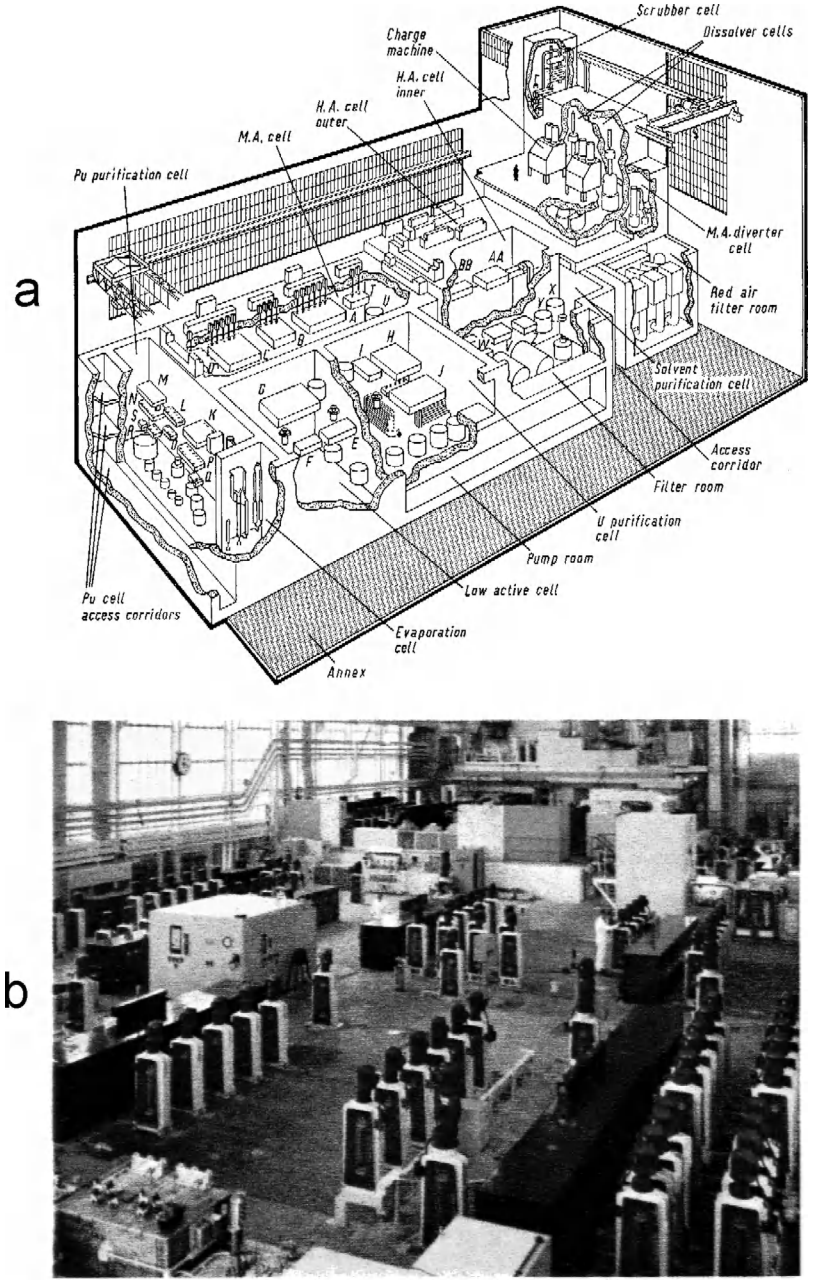


Figure 21.16 (a) General layout of BNFL Magnox reprocessing plant at Sellafield, UK. (b) Cell top of primary separation plant showing stirrer motors for mixer-settlers below.

Even such systems may fail and dual equipment is therefore sometimes installed in parallel cells. With the process running on the spare equipment, the failing equipment must be repaired. This requires efficient decontamination both on inside and outside of the equipment, provisions for which must be incorporated in the original design. This design requires dividing the plant in a large number of shielded cells so that the repair workers are protected from the radiation of the functioning plant.

Remote maintenance is more expensive, but may be safer for the personnel and desirable from the standpoint of continuity of operations because equipment replacements can be carried out quickly and interruption of operation is relatively brief. Modern robot technology simplifies such remote maintenance.

21.7. REPROCESSING OF THORIUM FUELS

The thorium-containing fuels of present interest are only those of the kind used in HTGR and HWR; in the future fuel from MSR-like[4] (or other) transmutation devices may become important. In the HTGR fuel elements the fertile ThO_2 and fissile $^{235}UO_2$ (or $^{233}UO_2$) particles are coated differently and embedded in a graphite matrix.

In case of the HTGR, the spent graphite fuel elements are mechanically crushed and burned to eliminate the graphite matrix and pyrolytic carbon coating from the fuel particles. Leaching permits separation of the fissile and the fertile particles because the fissile particles have a silicon carbide coating which remains intact during burning and leaching, while the all-pyrolytic carbon coatings on the fertile particles are burned away, allowing the oxide ash to be dissolved by a leach solution, consisting of HNO_3 and F^-. The solution is clarified and adjusted to proper acidity for solvent extraction (paragraph (i) below). The undissolved residue resulting from clarification is dried and classified for further treatment (paragraph (ii) below). The burner off-gas streams are passed through several stages of filtration, scrubbing, and chemical reaction to remove the entrained and volatile fission products, as well as ^{14}C containing CO_2, prior to atmospheric discharge.

In case of oxide fuel, it has to be cut and dissolved in a way similar to that described for uranium fuels.

(i) The *acid Thorex solvent extraction process* is used to purify and to separate the ^{233}U and the thorium. Three solvent extraction cycles are used. In the first, the uranium and thorium are coextracted by 30% tributyl phosphate (TBP) from $\sim$5 M HNO_3 and then stripped into an aqueous phase. In the second cycle, the uranium and thorium are separated by controlling the extraction conditions using $\leq$1 M HNO_3. The uranium is extracted and processed by an additional solvent extraction cycle for final purification, while the thorium remains in the aqueous raffinate stream. Following

[4] MSR = Molten Salt Reactor.